

# 3D Exoscopes in Microsurgery: How 3D Exoscopes Reconfigure the Workflow of Assistant Surgeons and Nurses in the Operating Room

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Figure 1: Different OR environments in (a) traditional heads-down microscopic surgery and (b) heads-up exoscopic surgery

## ABSTRACT

With advanced technologies, 3D exoscopes have been introduced in the operating room and considered as an alternative to microscopes which have been a gold standard visualization tool in microsurgery. While many recent papers have examined the impact of the 3D exoscopes on a surgeon's performance, it has not been investigated how the 3D exoscopes affect the dynamic socio-technical environment of the operating room. We present preliminary findings from a field study exploring how the 3D exoscopes reconfigure the existing practice of surgical team members (i.e., assistant surgeons, scrub nurses, and circulating nurses). We identified that one of the big advantages of the 3D exoscopes, the 3D visualization, rather impedes their workflow and teamwork. We note that design factors of the 3D exoscope should be carefully considered to build a better operating room environment for all surgical members.

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## CCS CONCEPTS

• **Human-centered computing** → Empirical studies in collaborative and social computing; • **General and reference** → Empirical studies.

## KEYWORDS

microsurgery; 3D exoscope; teamwork; surgical team; operating room nurses; assistant surgeons

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## 1 INTRODUCTION

While microscope systems have been used as a fundamental visualization tool in microsurgery over several decades, high definition three-dimensional (3D) exoscope systems have only been recently introduced as an alternative to the microscopes and considered as the future of microsurgery [11, 15]. The 3D exoscope system consists of a scope with a robotic arm, hand grips and a foot pedal to control the exoscopes, and a 3D screen where a 3D view of the surgical field is projected. It enables surgeons to perform surgeries

in a heads-up position which eases their neck and back pain and all surgical members to easily follow a surgical procedure by watching a 3D screen. Most research has investigated the impact of the 3D exoscopes on a primary surgeon's performance in terms of its technical (e.g., image quality, lighting, magnification) and surgical aspects (e.g., optics, ergonomics, maneuverability of instruments) [11, 13, 15]. Beyond the benefits to surgeons, this state-of-the-art technology is also expected to improve communication between surgical team members during surgery, because people in an operating room (OR) can easily follow a surgical procedure and have the same view with the primary surgeon by watching a 3D screen [3, 8]. While new technology not only shapes individual experience but also reconfigures team dynamics in the OR [4, 7, 9], we have not found any ethnographic studies exploring how the 3D exoscope would affect the socio-technical environments of the OR.

In this poster, we present preliminary results of our field study to investigate how the 3D exoscope system alters the existing practice of surgical team members (i.e., assistant surgeons, scrub nurses, and circulating nurses) who are not the primary users (i.e., a primary surgeon) but who closely collaborate with them in the OR. Drawing on observations and interviews, we investigate what concomitant effects were created by the new system (i.e., 3D exoscopes) and identify what design factors of the 3D exoscope should be carefully considered to facilitate teamwork and build a better OR environment for all surgical members.

## 2 BACKGROUND

### 2.1 Microscopes versus 3D Exoscopes

In microsurgery, it is critical to identify, handle with care, and dissect delicate neurovascular microstructures. It requires excellent magnification and illumination to enable the surgeons to examine the fine details of an operating field without causing damage to the tissue. In this regard, the microscopes have been used as “a gold standard visualization device [12]” in microsurgery since its introduction in the early 1960s. The conventional microscopes have, however, several limitations, including limited focal range, image quality, portability, and especially in terms of ergonomics [11, 15]. Surgeons stand and operate in uncomfortable positions during surgery, because they need to fix their eyes to the eyepieces of the microscopes [13, 15] (see Figure 1 (a)). Moreover, the surgeons use either a mouth switch or handgrips to constantly adjust the view of the microscopes [2].

On the other hand, the 3D digital exoscope systems –while utilizing powerful digital optical technologies– enable surgeons to perform surgeries in a heads-up position and thereby ease their neck and eventual back pain. In exoscope systems, surgeons wear 3D glasses to watch a 3D screen presenting the surgical field. They use a foot pedal or handgrips to control basic functions (e.g., focus, zoom, and light intensity) and to adjust the camera position [15]. Therefore, they do not need to fix their eyes to the binoculars and use the mouth switch. The 3D screen presenting the magnified surgical field is mostly located next to the operating table and the surgeon faces perpendicular to the 3D screen [16]. At the same time, all people in the OR can follow the surgical process by attending the

3D screen (see Figure 1(b)).<sup>1</sup> While there are many differences between microscope and 3D exoscope in their technical aspects (e.g., illumination, magnification, field of view), in this study, we report on the impacts of the 3D exoscope system on the OR environments and surgical members' workflow.

### 2.2 Roles of Surgical Team Members

A typical operating room is composed of multi-disciplinary team members as following:

- Primary attending surgeon: leads whole surgical procedures
- Assistant surgeon (AS): provides assistance to the primary surgeon
- Scrub nurse (SN): handles and passes surgical tools to the surgeon in the sterile field
- Circulating nurse (CN): helps all surgical members by managing activities that occurring in the non-sterile field
- Anesthesiologist: monitors a patient's vital sign and provides pain control throughout a surgical procedure
- Technician : helps the surgical team to set up surgical devices in the OR

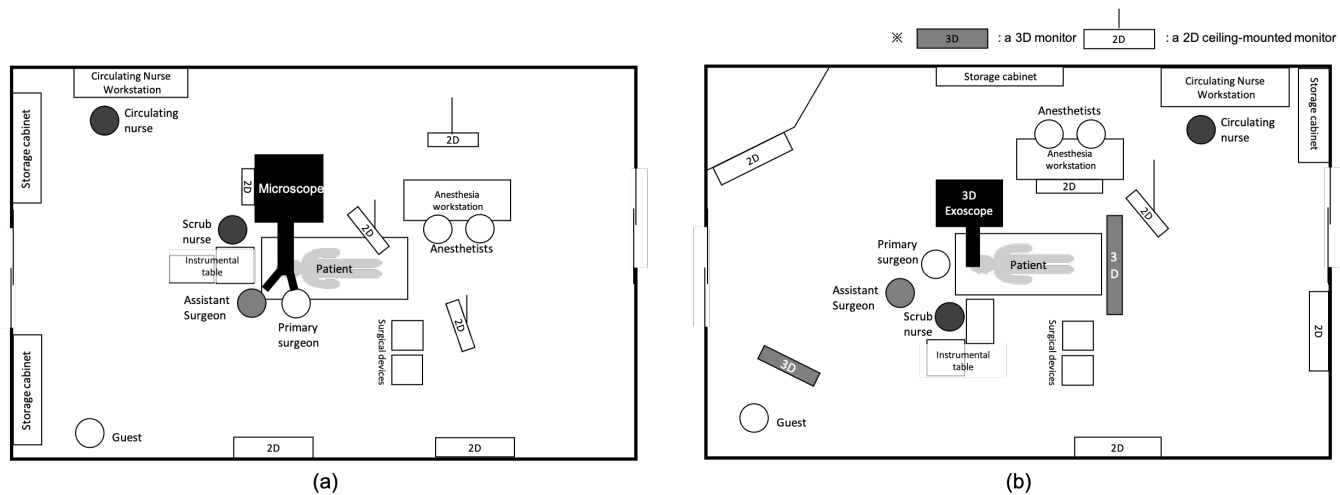
The surgical team members could be categorized into sterile and non-sterile members depending on their workspace in the OR. The surgeons and the scrub nurses only work in the sterile area, and they closely collaborated with each other during surgery. Main surgical procedures are done by the primary attending surgeon who are assisted by the scrub nurse and the assistant surgeon. When the critical elements of surgeries are done, the primary surgeon sometimes taps assistant surgeons to help at the end of surgery (e.g., wound closure). On the other hand, the circulating nurses, the anesthetists, and the technicians work in the non-sterile area. Among the non-sterile members, the circulating nurses actively collaborate with sterile members while work as a mediator between sterile and non-sterile members. Therefore, the movement pattern of the circulating nurse is the most dynamic as they travel various zones (e.g., storage, computer workstation) within and outside the OR [5]. While they are required to provide surgical supplies to the surgeons and the scrub nurses, they are also responsible for documentation of all supplies in the OR at their workstation and monitoring the whole situation in the OR. The circulating nurses and the scrub nurses all have the same expertise and change roles every other surgery. In this study, we only focus on assistant surgeons, scrub nurses, and circulating nurses who are not the main operator of the surgical systems, microscopes and 3D exoscopes, but who assist and closely collaborate with the main operator, the primary surgeon, during surgery. This is to understand how the new system, the 3D exoscope, reconfigure the team dynamics in the OR.

## 3 METHODS

### 3.1 Observations

By observing surgeries under two different OR settings (3D exoscopes and microscopes) at two major hospitals in Finland (from

<sup>1</sup>Although there is also an advanced microscope with a 3D screen where surgical members can share the surgical moment by watching the screen just like the 3D exoscope, the surgeon still has to look the surgical field through the eyepiece.



**Figure 2: Arrangement of the operating room for (a) microscope and (b) 3D exoscope in brain surgery (Note: the position of surgical devices and sterile members are changed depending on surgical procedure)**

here on referred to as Hospital A and B), we aimed to gain a better understanding of microsurgical practice with the microscopes and to explore how surgical workflow and teamwork would be reshaped by the 3D exoscopes. We observed surgical procedures (i.e., spine and brain surgery) using the 3D exoscopes (four cases at Hospitals A and B, approximately 15.5 hours) and microscope (four cases at Hospital B, approximately 9.5 hours). This data allows us not to compare two hospitals but to understand surgical environments with the microscope and 3D exoscope in general. When we observed the surgery, both hospitals were at the early stage of adopting the 3D exoscope. During the observation, the first author wrote field notes, focusing on the type of tools the surgical team used, interactions within surgical team members (i.e., a primary surgeon, assistant surgeons, scrub nurses, circulating nurse, and anesthetists) the time-stamped notes of interesting events, and OR environments (see Figure 2).

### 3.2 Interview

We interviewed two assistant surgeons and three OR nurses who have been experienced in both microscopes and 3D exoscopes. When we interviewed the participants, all of them considered microscopic surgery as routine work and they were at the early stage of using 3D exoscopes. The main focus of the interviews was to understand how the 3D exoscopes reshape OR environments where the microscopes have been mainly used. We started by asking surgical team members to share their experience with microscopes and 3D exoscopes. Then, we specifically asked questions to understand how the 3D exoscope changes their work practices in terms of the following: OR setups, their roles and responsibilities, and team collaboration in the OR. When we interviewed the OR nurses, we tried to understand whether and how they would differently perceive the challenges that scrub nurses and circulating nurses would have. At the end of the interviews, we asked them what should be considered to build a better OR environment with the 3D exoscopes.

Interviews were done either in-person or via video call and lasted from 20 minutes to 60 minutes.

### 3.3 Analysis

We used the field notes to further understand the OR environments with either microscopes or 3D exoscopes. We iteratively coded and analyzed the interviews through a grounded theory approach [17]. Through the coding process, we examined how the 3D exoscope alters existing practice of surgical team members during surgery.

## 4 FINDINGS

### 4.1 From Assistant Surgeons' perspectives

**4.1.1 Educational value.** With the 3D exoscopes, assistant surgeons were able to share the same surgical view with the primary surgeon and to easily check 3D anatomical details. Thanks to the heads-up approach, they could follow the surgery in a comfortable posture (*"It is definitely helpful for us to follow surgical procedures [Assistant Surgeon (AS)1]."*)

**4.1.2 3D visualization.** However, the 3D learning environment was not always ensured for the assistant surgeons as they sometimes stood perpendicular to the primary surgeon, not to the 3D screen. There were other issues reported regarding the 3D visualization when the assistant surgeon took the main operator role to perform wound closure at the end of surgery. It was not easy for them to set up the best work environment where they could fully benefit from 3D exoscopic visualization. They pinpointed that they could easily change the working distance and the height of the microscope by themselves. In 3D exoscope settings, they needed other surgical members help to adjust the height of the 3D screen, but they did not have enough time to do that.

**4.1.3 Foot pedal.** They were not familiar with the foot pedal of 3D exoscope as it is not commonly used in microsurgery. During the observations, we found several moments when assistant surgeons

struggled to use it. In this case, they tended to use hand grips to control the position of the camera which could be possible reasons for a longer operation time. However, they did not consider the foot pedal to be a big challenge to use the exoscope and mentioned that they just needed more time to practice.

## 4.2 From Nurses' Perspectives

**4.2.1 Educational value.** All nurses noted that the 3D exoscope enables them to understand the surgery better as they can see the anatomical details by checking the 3D screen.

**4.2.2 Communication between the Primary Surgeon and the Scrub Nurse.** They also highlighted that the surgeon's ergonomic postures and increased freedom of movement improved scrub nurses' work environments too: *"It's not like the microscopes. The surgeon and the scrub nurse both can be in an ergonomic posture and look each other [Nurse (N) 1]."* The scrub nurse can easily pass tools to the surgeons in the 3D exoscope setting and communicate better with the surgeon: *"When the surgeons use the microscopes, they are just in that world [N1]."* *"Surgeons bite the mouthpiece. so hard to understand what they're saying [N3]."*

**4.2.3 3D visualization.** Although the 3D screen only provides a better visualization when people wear 3D glasses, the scrub nurse did not prefer to wear them: *"Many nurses, include me, get nauseous ... because we have to switch attention all the time, looking at the screen and the instrument table [N2]."* As they could not solely watch the 3D screen, the 3D visualization rather interrupted their workflow. Therefore, they mostly watched a normal 2D screen: *"We (nurses) don't always have such nice screens in all the OR. Sometimes there's like a small screen which is further away, where you have to look how the surgery is going [N1]."* In this regard, the scrub nurse actually did not get benefits from the 3D visualization and their task was rather hindered by the 3D screen.

**4.2.4 Communication between the Circulating Nurse and the Scrub Nurse.** The size and the location of the 3D screen also disrupted the collaboration between the scrub nurse and the circulating nurse. They usually communicate with each other by using facial expressions or gestures in the microscope setting. In the exoscope setting, the 3D screen was mostly located between the circulating nurse workstation and the scrub nurse. Therefore, it blocked their field of view. For the circulating nurses, it made hard to be alert to the needs of the surgical team. They had to have longer travel paths between their workstation or storage cabinet and the sterile field. From the scrub nurse's perspective, they had to loudly call the circulating nurse to ask for help which would interrupt the surgeon. Therefore, they tried to minimize this occasion and it made it difficult for them to perform their work: *"You (scrub nurses) would not want to bother them (surgeons) unless you really, really need to. Then it makes your work kind of more difficult [N1]."* When we asked if the surgical team discussed the OR setup for the exoscope, they said that the surgeons mainly determined the whole setup. They mostly got directions from the surgeons in the OR. To some extent, they perceived that OR setup would be mainly designed for the main operator, the primary surgeon, because the patient safety should be a priority in the OR: *"We are usually not the first [N2]."* However, the OR nurses considered that the current challenges

were natural because the 3D exoscope surgery was not routine work for them yet. They felt they just needed more experience to get familiar with the new environment, similarly as what the assistant surgeons mentioned.

## 5 CONCLUSIONS

We present preliminary findings from a field study exploring how the 3D exoscopes reconfigure the work practice of assistant surgeons and nurses in the OR. We note that the issues related to the 3D exoscope are interwoven in various levels: technical, individual, and interpersonal/team levels. One of the big advantages of 3D exoscope, the 3D visualization, impedes the surgical workflow and teamwork. This was mainly because the OR environments were established not for them but for the primary attending surgeon who directly operates surgery and controls surgical devices. The rest of the surgical team members had to adjust to the new environment driven by the 3D exoscope. However, our participants did not substantially worry about the changes driven by the 3D exoscope, and they were convinced that they would overcome the challenges when they have more experience. This seems somehow a part of culture and clinical professionalism. However, it could be a barrier to shorten learning curves which is necessary to overcome to promote surgical performance (i.e., individual and team performance) and outcomes. Rather than forcing the secondary users to adjust themselves to the system, one would argue that there is a need to further understand how new surgical technologies shape a socio-technical process (e.g., communication, teamwork) in the OR [1, 9, 14]. This would contribute to designing usable, safe, and effective interventions in the OR which would be beneficial to all surgical members and patients in the long run.

Due to the limited number of participants in this study, we are planning to perform more extended observations and interviews to identify further design implications of 3D exoscope that affect surgical members' tasks and workflow in the OR. By applying activity theory [6, 10], we will clarify how the new surgical system reconfigures the previous activity system with the microscope and identify how it alters the rules and division of labors in the OR. To do so, we aim to present design guidelines of OR environments for the 3D exoscope systems that would be beneficial for facilitating the transition of OR environments from microscopes to 3D exoscopes and for developing technologies that can support better teamwork in the OR.

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